

# Miguel Aréjula Aísa

arejula10@gmail.com | +34 601 49 10 89 | LinkedIn | GitHub | are-dev.es

## Profile

Software Engineering Master's student at SDU and former intern at MongoDB (Industry Solutions). Experienced in full-stack development, prototyping, and building practical AI agents, tool workflows, and LLM integrations using TypeScript, Next.js, and Python.

## Experience

### Industry Solutions Intern, MongoDB

July 2026 - September 2026

- Built an offline-first wallet demo for Financial Services: an on-device ObjectBox store synced bidirectionally with MongoDB Atlas, payments queued offline and replayed on reconnect, semantic transaction search on Voyage AI embeddings, and a LangGraph assistant on a local Ollama model.
- Maintained and extended the Next.js demo-deployment tool used across the Industry Solutions team, shipping bug fixes and new features to production.
- Wrote a Claude Code skill that drafts a full MongoDB Solutions Library article from a solution repository, encoding the team's template, taxonomy, and style rules so every write-up starts consistent instead of from a blank page.
- Shared the solution externally through a blog post, a MongoDB Solutions Library entry, and a YouTube video.

### Teaching Assistant, SDU

February 2026 - May 2026

- Support students in understanding core software architecture concepts by assisting with lab sessions, assignments, and problem-solving, while clearly explaining technical concepts and collaborating with professors.

### Software Engineer, University of Zaragoza

June 2024 – August 2025

- Led end-to-end development of a clinical web app for Hospital Clínico Lozano Blesa, covering medical stakeholder requirements gathering, UI prototyping in Figma, and system delivery.
- Developed a full-stack system using React and PostgreSQL, with RESTful APIs in Express and FastAPI, streamlining operations for ~50 patients/day.
- Integrated structured clinical data into a relational model, validating clinical pathways through Petri nets and formal methods.

## Projects

### Leafy Wallet

[github.com/leafy-wallet](https://github.com/leafy-wallet)

- Offline-first personal wallet demo for Financial Services: an on-device ObjectBox store syncing bidirectionally with MongoDB Atlas, with payments queued offline and replayed on reconnect.
- Implemented multi-document ACID transactions, reconciliation against an external payment platform (PSP) behind an OAuth PKCE SSO flow, and semantic transaction search on Voyage AI embeddings (Atlas \$rankFusion online, on-device HNSW offline).
- Built a LangGraph assistant on a local Ollama model, routing questions to MongoDB MCP tools online and to the on-device store offline. **Tools Used:** Next.js, FastAPI, Python, MongoDB Atlas, ObjectBox, LangGraph, Ollama, Docker, Figma.

### AgroNet

[github.com/AgroNet](https://github.com/AgroNet)

- Designed and developed a full-stack collaborative web platform for farmers, integrating real-time market prices, personalized weather alerts, and interactive geospatial data visualization across Spain.
- Managed communication with the backend hosted on AWS, ensuring reliable API integration and real-time data synchronization. **Tools Used:** Astro, Node.js, TypeScript, Figma, MongoDB, AWS, Vercel.

### ByronHUB

[github.com/ByronHUB](https://github.com/ByronHUB)

- Developed ByronHub, a space booking application for the Ada Byron college building, using a four-tier hexagonal architecture with Domain-Driven Design principles.
- Containerized the system with Docker Compose, deployed it to AWS and Railway with continuous deployment from GitHub, and covered it with Postman and Jest unit/integration tests in GitHub Actions pipelines (>80% coverage). **Tools Used:** Node.js, TypeScript, React, PostgreSQL, RabbitMQ, Docker, AWS, Railway, Jest, GitHub Actions, Figma.

## Skills & Tools

**Languages & Core:** TypeScript, JavaScript, Python, Go, C/C++, Java, SQL, HTML5, CSS3.

**UI/UX & Frontend:** React, Next.js, Astro, Tailwind CSS, Figma.

**AI & Agentic Workflows:** AI Agents (LangGraph, MCP tools), Tool Calling, Claude Code Skills, LLM Integration (Ollama), Vector Search.

**Backend, DB & Cloud:** FastAPI, Node.js, Express, MongoDB Atlas, PostgreSQL, Docker, AWS, Vercel, REST APIs, Git.

## Education

### Master in Software Engineering, University of Southern Denmark

Sept 2025 – June 2027

### Bachelor in Software Engineering, University of Zaragoza

Sept 2021 – June 2025

- Achieved high honors in Information Systems II and Verification and Validation, demonstrating exceptional proficiency and understanding in these subjects.